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Lighweight vandal resistant parking meter

Abstract

A parking meter may have a housing that is formed at least significantly of an impact resistant material. In particular, the impact resistant material may comprise one of a polycarbonate copolymer or polycarbonate blended resin, a nylon-based copolymer, or a nylon-based polymer that includes long glass fiber or Kevlar fiber. The housing encloses the parking meter components to provide protection for the parking meter components against an external environment and vandalism attacks.

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D454421	12/2001	Jeon et al.	D99/28	N/A
D454807	12/2001	Cappiello	D10/109	N/A
6354425	12/2001	Tuxen et al.	194/350	N/A
D456587	12/2001	Kit et al.	D99/28	N/A
6366220	12/2001	Elliott	340/928	N/A
6373401	12/2001	Ho	340/932.2	N/A
6373422	12/2001	Mostafa	341/155	N/A
6373442	12/2001	Thomas et al.	343/767	N/A
6380851	12/2001	Gilbert et al.	340/517	N/A
D460005	12/2001	Jacquet	D10/42	N/A
D461728	12/2001	Tuxen et al.	D10/42	N/A
6456491	12/2001	Flannery et al.	361/685	N/A
D463749	12/2001	Petrucelli	D10/109	N/A
6457586	12/2001	Yasuda et al.	209/534	N/A
6467602	12/2001	Bench et al.	194/217	N/A
6477505	12/2001	Ward, II et al.	N/A	G06F 17/60
6477875	12/2001	Field et al.	70/491	N/A
D467954	12/2001	Suzuki et al.	D16/208	N/A
6493676	12/2001	Levy	705/13	N/A
6505774	12/2002	Fulcher et al.	N/A	N/A
D471238	12/2002	Showers et al.	D20/8	N/A
D472362	12/2002	Zerman et al.	D99/28	N/A
6527172	12/2002	Lewis et al.	N/A	G06F 17/60
6539370	12/2002	Chang et al.	707/2	N/A
6559776	12/2002	Katz	340/932.2	N/A
6575281	12/2002	Lee	192/49	N/A
D477030	12/2002	Kolls et al.	D20/8	N/A
D481516	12/2002	Magee et al.	D99/28	N/A
D485417	12/2003	Magee et al.	D99/28	N/A
6697730	12/2003	Dickerson	701/200	N/A
D488280	12/2003	Zerman et al.	D99/28	N/A
6731942	12/2003	Nageli	455/458	N/A
D490727	12/2003	Kido et al.	D10/60	N/A
D492080	12/2003	Magee et al.	D99/28	N/A
D492081	12/2003	Magee et al.	D99/28	N/A
D492085	12/2003	Korte et al.	D99/43	N/A
6747575	12/2003	Chauvin et al.	340/932.2	N/A
6763094	12/2003	Conn et al.	379/91.01	N/A
D494730	12/2003	Magee et al.	D99/28	N/A
6791473	12/2003	Kibria et al.	340/932.2	N/A
RE38626	12/2003	Kielland	340/937	N/A

D497393	12/2003	Herbst	D20/99	N/A
6799387	12/2003	Pippins	37/456	N/A
D497814	12/2003	Odinotski et al.	D10/42	N/A
D498795	12/2003	Nunn	D20/8	N/A
6812857	12/2003	Kassab et al.	340/932.2	N/A
6823317	12/2003	Ouimet et al.	705/13	N/A
6856922	12/2004	Austin et al.	702/63	N/A
6874340	12/2004	Berman	7/161	N/A
6885311	12/2004	Howard et al.	340/932.2	N/A
D505240	12/2004	Swaine et al.	D99/28	N/A
D505765	12/2004	Swaine et al.	D99/28	N/A
6889899	12/2004	Silberberg	235/384	N/A
D506509	12/2004	Nunn	D20/9	N/A
D506769	12/2004	Asai	D16/203	N/A
6914411	12/2004	Couch et al.	320/101	N/A
D508064	12/2004	Ramirez	D16/200	N/A
6929179	12/2004	Fulcher et al.	235/381	N/A
6946974	12/2004	Racunas Jr.	340/932.2	N/A
D510751	12/2004	Magee et al.	D20/3	N/A
7004385	12/2005	Douglass	235/379	N/A
7014355	12/2005	Potter, Sr. et al.	368/90	N/A
7019420	12/2005	Kogan et al.	307/150	N/A
7019670	12/2005	Bahar	340/932.2	N/A
7023360	12/2005	Staniszewski	340/932.2	N/A
7027773	12/2005	McMillin	455/41.2	N/A
7027808	12/2005	Wesby	455/419	N/A
7028888	12/2005	Laskowski	235/379	N/A
7029167	12/2005	Mitschele	368/90	N/A
D520883	12/2005	Hillard et al.	D10/50	N/A
7040534	12/2005	Turocy et al.	235/379	N/A
7046123	12/2005	Goldberg	340/309.8	N/A
D524834	12/2005	Dozier et al.	D16/200	N/A
7071839	12/2005	Patel et al.	340/933	N/A
7104447	12/2005	Lopez et al.	235/384	N/A
7110762	12/2005	Cameron et al.	455/434	N/A
D530880	12/2005	Magee et al.	D99/28	N/A
7114651	12/2005	Hjelmvik	235/384	N/A
7131576	12/2005	Utz et al.	235/379	N/A
D535268	12/2006	Holger	D14/130	N/A
7181426	12/2006	Dutta	705/37	N/A
7183999	12/2006	Matthews et al.	345/1.1	N/A
D538505	12/2006	Kang et al.	D99/28	N/A
7188070	12/2006	Dar et al.	750/1	N/A
D543588	12/2006	Herbert et al.	D20/8	N/A
7222031	12/2006	Heatley	N/A	N/A
7222782	12/2006	Lute, Jr. et al.	235/379	N/A
7230545	12/2006	Nath et al.	340/932.2	N/A
7237176	12/2006	Briggs et al.	714/67	N/A
D546365	12/2006	Jost et al.	D16/221	N/A
7237716	12/2006	Silberberg	235/384	N/A
7253747	12/2006	Noguchi	340/932.2	N/A
7262737	12/2006	Zarnowitz et al.	343/702	N/A
7284692	12/2006	Douglass	235/379	N/A
D559792	12/2007	Gemme et al.	D13/164	N/A
7319974	12/2007	Brusseaux	705/13	N/A
7321317	12/2007	Nath et al.	340/933	N/A
7330131	12/2007	Zanotti et al.	340/932.2	N/A
D570920	12/2007	Choi	D20/2	N/A
7382238	12/2007	Kavaler	340/438	N/A
7382281	12/2007	Kavaler	340/941	N/A
7382282	12/2007	Kavaler	340/941	N/A
7385484	12/2007	Nath et al.	340/426.1	N/A
7388349	12/2007	Elder et al.	320/104	N/A

7388517	12/2007	Kavaler	340/941	N/A
7391339	12/2007	Howard et al.	340/932.2	N/A
7393134	12/2007	Mitschele	368/90	N/A
D575168	12/2007	King et al.	D10/42	N/A
7419091	12/2007	Scanlon	235/379	N/A
7427931	12/2007	Kavaler et al.	340/941	N/A
D579795	12/2007	Martinez et al.	D10/42	N/A
7445144	12/2007	Schlabach et al.	235/379	N/A
D582125	12/2007	Kang et al.	D99/28	N/A
D587141	12/2008	King et al.	D10/42	N/A
D588916	12/2008	DeMarco	D9/504	N/A
D591181	12/2008	Kanbar	D10/40	N/A
D592825	12/2008	Kang et al.	D99/28	N/A
7533802	12/2008	McGinley et al.	235/379	N/A
7533809	12/2008	Robinson et al.	N/A	G06K 5/00
7554460	12/2008	Verkleeren et al.	340/870.02	N/A
D596373	12/2008	Kang et al.	D99/28	N/A
D596374	12/2008	Kang et al.	D99/28	N/A
7558564	12/2008	Wesby	455/419	N/A
7575166	12/2008	McNamara	235/440	N/A
7579964	12/2008	Nath et al.	340/933	N/A
D600152	12/2008	Stack et al.	10/5	N/A
7594235	12/2008	Moreau	719/310	N/A
D602225	12/2008	Lute et al.	D99/28	N/A
D604189	12/2008	Leer et al.	D10/123	N/A
7611045	12/2008	Lute et al.	235/379	N/A
7617120	12/2008	Derasmo et al.	705/13	N/A
D605145	12/2008	Kellar	D14/125	N/A
7624919	12/2008	Meek et al.	235/379	N/A
7632419	12/2008	Grimbergen et al.	216/60	N/A
7671803	12/2009	Neill et al.	343/700MS	N/A
7679526	12/2009	Mardirossian	340/932.2	N/A
7699224	12/2009	Molé	235/439	N/A
D615274	12/2009	Kim et al.	D99/28	N/A
7717327	12/2009	Gomes	235/379	N/A
D617799	12/2009	Odhe et al.	D14/420	N/A
7726558	12/2009	Lute et al.	235/379	N/A
7739000	12/2009	Kevaler	701/1	N/A
D620050	12/2009	Hultzman et al.	D20/2	N/A
7748620	12/2009	Gomez et al.	235/383	N/A
7772720	12/2009	McGee et al.	307/66	N/A
7780072	12/2009	Lute et al.	235/379	N/A
7783530	12/2009	Stemmer et al.	750/28	N/A
D624274	12/2009	Pendleton	D99/28	N/A
D625305	12/2009	Bleck et al.	D14/307	N/A
7806248	12/2009	Hunter et al.	194/317	N/A
D627814	12/2009	Tzeng et al.	D16/202	N/A
7825826	12/2009	Welch	340/932.2	N/A
D629585	12/2009	Bleck et al.	D99/28	N/A
7854310	12/2009	King et al.	194/350	N/A
7855661	12/2009	Ponert	340/932.2	N/A
7874482	12/2010	Mitschele	235/384	N/A
D632740	12/2010	Adeline	D20/2	N/A
7891546	12/2010	Steinbach et al.	235/375	N/A
D634417	12/2010	Abbondanzio	D23/366	N/A
D635035	12/2010	Matsumoto	D10/41	N/A
7899583	12/2010	Mendelson	701/1	N/A
7908149	12/2010	Dar et al.	705/1	N/A
7933841	12/2010	Schmeyer et al.	705/64	N/A
7945670	12/2010	Nakamura et al.	N/A	G06F 15/173
D645223	12/2010	Bleck et al.	D99/28	N/A
8035533	12/2010	Kavaler	340/933	N/A
8044139	12/2010	Youn et al.	524/606	N/A

D651784	12/2011	Rohan et al.	D99/28	N/A
D652329	12/2011	MacKay et al.	D10/42	N/A
D652601	12/2011	Rohan et al.	D99/28	N/A
D652602	12/2011	Rohan et al.	D99/28	N/A
D652872	12/2011	Mougin et al.	D20/1	N/A
D653012	12/2011	Rohan et al.	D99/28	N/A
D653014	12/2011	Rohan et al.	D99/28	N/A
D653420	12/2011	Rohan et al.	D99/28	N/A
D653421	12/2011	Rohan et al.	D99/28	N/A
D653422	12/2011	Rohan et al.	D99/28	N/A
D653423	12/2011	Rohan et al.	D99/28	N/A
D653424	12/2011	Rohan et al.	D99/28	N/A
D654816	12/2011	MacKay et al.	D10/42	N/A
RE43245	12/2011	Ouimet et al.	705/13	N/A
D655263	12/2011	Tsai	D14/125	N/A
D656046	12/2011	Mackay et al.	D10/42	N/A
8138950	12/2011	Leung	340/932.2	N/A
8144034	12/2011	Kavaler	340/941	N/A
D659557	12/2011	Jones et al.	D10/42	N/A
8181857	12/2011	Lute et al.	235/379	N/A
8184019	12/2011	Chauvin et al.	340/932.2	N/A
D661603	12/2011	MacKay et al.	D10/42	N/A
D661608	12/2011	Kimmich	D10/98	N/A
8193540	12/2011	Huang et al.	257/81	N/A
8207394	12/2011	Feldkamp et al.	604/361	N/A
8232894	12/2011	Weiss et al.	340/932.2	N/A
8264401	12/2011	Kavaler	342/128	N/A
8279107	12/2011	Krstanovic et al.	342/27	N/A
D672525	12/2011	Lee et al.	D99/28	N/A
8325057	12/2011	Salter	N/A	G08B 23/00
D674985	12/2012	Lee	D99/28	N/A
D677035	12/2012	Lee et al.	D99/28	N/A
D677714	12/2012	Helgesson et al.	D18/4.4	N/A
8395532	12/2012	Chauvin et al.	340/932.2	N/A
8402281	12/2012	Dahl	718/189	N/A
8402380	12/2012	Kikin-Gil et al.	N/A	G06F 3/048
D680156	12/2012	Hernandez et al.	D18/4.4	N/A
8417715	12/2012	Bruckhaus et al.	707/758	N/A
D681717	12/2012	Helgesson et al.	D18/4.6	N/A
D682504	12/2012	Lee et al.	D99/28	N/A
8450627	12/2012	Mittleman et al.	200/600	N/A
D683779	12/2012	Uemachi et al.	D16/202	N/A
D683794	12/2012	Randall	D20/8	N/A
D683795	12/2012	Randall	D20/8	N/A
D684745	12/2012	Reuter et al.	D99/28	N/A
D684870	12/2012	Jones et al.	D10/42	N/A
8479909	12/2012	King et al.	194/302	N/A
8487754	12/2012	Mitschele	340/539.1	N/A
D687809	12/2012	Bergmann et al.	D14/209.1	N/A
8513832	12/2012	Hunter et al.	307/66	N/A
D689554	12/2012	Echanove Hernandez	D20/2	N/A
D690771	12/2012	Randall	D20/9	N/A
D691165	12/2012	Attoma	D14/486	N/A
8566159	12/2012	King et al.	705/14.53	N/A
8568055	12/2012	Marques Lito Velez Grilo	N/A	F16C 11/00
D692784	12/2012	Anderssen et al.	D10/70	N/A
D693983	12/2012	Budde et al.	D99/28	N/A
8573484	12/2012	Irudayam et al.	235/379	N/A
8589216	12/2012	Yu et al.	705/13	N/A
8590687	12/2012	King et al.	194/350	N/A
8595054	12/2012	King et al.	705/13	N/A
D696484	12/2012	Lee et al.	D99/28	N/A
D696485	12/2012	Lee et al.	D99/28	N/A

8621245	12/2012	Shearer et al.	713/300	N/A
8631921	12/2013	Jones et al.	194/350	N/A
8662279	12/2013	Jones et al.	194/350	N/A
8666801	12/2013	Cho	705/13	N/A
D702913	12/2013	Budde et al.	D99/28	N/A
D702914	12/2013	Budde et al.	D99/28	N/A
8684158	12/2013	Jones et al.	194/215	N/A
8710798	12/2013	Turner	320/109	N/A
D705090	12/2013	MacKay et al.	D10/42	N/A
8727207	12/2013	Church et al.	235/33	N/A
D707140	12/2013	King et al.	D10/42	N/A
D707141	12/2013	King et al.	D10/42	N/A
D707142	12/2013	King et al.	D10/42	N/A
D707574	12/2013	Ku et al.	D10/40	N/A
8749403	12/2013	King et al.	340/932.2	N/A
8770371	12/2013	MacKay et al.	194/350	N/A
8770372	12/2013	Dobbins et al.	194/350	N/A
8807317	12/2013	MacKay et al.	194/350	N/A
D714165	12/2013	Salama	D10/42	N/A
D715661	12/2013	McKaughan et al.	D10/53	N/A
D716156	12/2013	Jones et al.	D10/42	N/A
D716515	12/2013	Cha et al.	D99/28	N/A
8862494	12/2013	King et al.	705/13	N/A
8866624	12/2013	Ales et al.	340/604	N/A
D716671	12/2013	Jones et al.	D10/42	N/A
8884785	12/2013	Groft et al.	N/A	G06Q 20/3278
8936505	12/2014	Bengtsson	N/A	G07D 1/00
D722740	12/2014	Budde et al.	D99/28	N/A
8957787	12/2014	Al-Hmalan et al.	N/A	G08G 1/0175
D724290	12/2014	Cha et al.	D99/28	N/A
D724811	12/2014	Reitingner	D99/28	N/A
D724812	12/2014	Cha et al.	D99/28	N/A
D725504	12/2014	McEneany et al.	D10/42	N/A
D728190	12/2014	Sheley	D99/28	N/A
9002723	12/2014	King et al.	340/932.2	N/A
D732263	12/2014	Budde et al.	D99/28	N/A
9047712	12/2014	King et al.	705/1.1	N/A
9064358	12/2014	Lidror	N/A	H04Q 5/22
D733585	12/2014	Jones et al.	D10/42	N/A
D735437	12/2014	Randall	D99/28	N/A
D735438	12/2014	Randall	D99/28	N/A
D735439	12/2014	Randall	D99/28	N/A
D735440	12/2014	Randall	D99/28	N/A
D735963	12/2014	Randall	D99/28	N/A
D735964	12/2014	Randall	D99/28	N/A
D735965	12/2014	Randall	D99/28	N/A
D739637	12/2014	Randall	D99/28	N/A
D739638	12/2014	Randall	D99/28	N/A
D739639	12/2014	Randall	D99/28	N/A
9123184	12/2014	Jones et al.	N/A	G07B 15/00
9127964	12/2014	Schwarz et al.	340/943	N/A
9147345	12/2014	Agrait et al.	N/A	G08G 1/14
D741040	12/2014	Budde et al.	D99/28	N/A
D742090	12/2014	Randall	D99/28	N/A
9159080	12/2014	Fiorucci et al.	N/A	G06Q 30/0226
9169626	12/2014	Guler et al.	N/A	E03D 3/06
D742371	12/2014	Bopp et al.	D14/336	N/A
D743956	12/2014	Kraft et al.	D14/307	N/A
9196097	12/2014	Jones et al.	340/870.02	N/A
9196161	12/2014	Lai et al.	N/A	B06Q 1/48
D746015	12/2014	King et al.	D99/28	N/A
D746704	12/2015	Jones et al.	D10/42	N/A
D748364	12/2015	King et al.	D99/28	N/A

D748888	12/2015	King et al.	D99/28	N/A
D749000	12/2015	King et al.	D10/42	N/A
D749290	12/2015	Pollmann	D99/28	N/A
9262915	12/2015	Clem et al.	N/A	G08G 1/0112
9269216	12/2015	Keilwert	N/A	G07F 17/3206
D750513	12/2015	King et al.	D10/70	N/A
D750978	12/2015	van Slooten et al.	D10/41	N/A
D752552	12/2015	D'Ascanio et al.	D14/216	N/A
D755649	12/2015	King et al.	D10/42	N/A
D755650	12/2015	King et al.	D10/42	N/A
D756807	12/2015	King et al.	D10/42	N/A
D756808	12/2015	King et al.	D10/42	N/A
D762040	12/2015	King et al.	D99/28	N/A
9384554	12/2015	Xu et al.	N/A	G06T 7/0075
9391474	12/2015	Hunter et al.	N/A	H02J 9/06
9406056	12/2015	MacKay et al.	N/A	G06Q 30/02
9407608	12/2015	Mullick et al.	N/A	G06F 17/00
9424691	12/2015	King et al.	N/A	G07B 15/02
9437050	12/2015	Hilton et al.	N/A	G07B 15/00
9443236	12/2015	MacKay et al.	N/A	G06Q 20/18
D769570	12/2015	Kraft et al.	D99/28	N/A
D769571	12/2015	Lee et al.	D99/28	N/A
D769572	12/2015	Park et al.	D99/28	N/A
9489776	12/2015	Keller et al.	N/A	G07B 15/02
9494922	12/2015	MacKay et al.	N/A	G04C 11/026
9508198	12/2015	King et al.	N/A	G07B 15/02
9508260	12/2015	Shaik	N/A	G08G 1/14
9519761	12/2015	Jakobsson	N/A	G06F 7/04
9524498	12/2015	Jones et al.	N/A	G06Q 20/145
9536235	12/2016	Jones et al.	N/A	G06Q 20/145
9536370	12/2016	Jones et al.	N/A	G07F 17/246
D779771	12/2016	King et al.	D99/28	N/A
D779772	12/2016	King et al.	D99/28	N/A
D779773	12/2016	King et al.	D99/28	N/A
D779774	12/2016	King	D99/28	N/A
9580198	12/2016	Ehrmann et al.	N/A	B23B 41/00
D781024	12/2016	King et al.	D99/28	N/A
D782349	12/2016	Konotopskyi et al.	D10/70	N/A
9661403	12/2016	King et al.	N/A	H04Q 9/00
9685027	12/2016	King et al.	N/A	G07F 17/248
9692256	12/2016	Hunter et al.	N/A	H02J 9/061
D791621	12/2016	Loye et al.	D10/42	N/A
D792049	12/2016	Oh et al.	D99/28	N/A
9697506	12/2016	Jones et al.	N/A	G06Q 20/145
9707911	12/2016	Myers et al.	N/A	B60R 16/037
D795523	12/2016	King et al.	D99/28	N/A
9728085	12/2016	Schwarz et al.	N/A	G08G 1/017
9746954	12/2016	Wyatt et al.	N/A	G06F 3/0416
9749823	12/2016	Rowe et al.	N/A	H04W 4/18
D796355	12/2016	Cho et al.	D10/70	N/A
9773351	12/2016	Hudson et al.	N/A	G07B 15/02
D799992	12/2016	Ehara	D10/41	N/A
D801304	12/2016	Kiafoulis et al.	D14/214	N/A
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9779565	12/2016	Rabbat	N/A	G07C 9/00
9805518	12/2016	King et al.	N/A	G07B 15/02
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9842346	12/2016	Fiorucci et al.	N/A	G06Q 30/0226
9842455	12/2016	MacKay et al.	N/A	G07F 7/10

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9934645	12/2017	MacKay et al.	N/A	G07F 17/24
10069233	12/2017	Goto	N/A	H01R 13/4223
D832553	12/2017	Angelopoulos et al.	D99/28	N/A
10089814	12/2017	King et al.	N/A	G07F 17/0021
D832809	12/2017	Hardi	D14/204	N/A
10141629	12/2017	Mackay	N/A	G04C 11/026
D834971	12/2017	Ahn et al.	D10/70	N/A
10154029	12/2017	Griffin	N/A	H04L 63/0861
10192388	12/2018	MacKay et al.	N/A	G06F 17/24
D843973	12/2018	Klosowski et al.	D14/216	N/A
10262345	12/2018	King et al.	N/A	G06Q 30/0284
10275650	12/2018	Laaksonen	N/A	G06K 9/00
10297150	12/2018	Schwarz et al.	N/A	G08B 23/00
10299018	12/2018	King et al.	N/A	H04Q 9/00
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10315665	12/2018	Halder	N/A	G01C 22/00
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D854430	12/2018	Elrod et al.	D10/52	N/A
10366546	12/2018	King et al.	N/A	G06Q 10/00
D858381	12/2018	Ahn et al.	D12/173	N/A
10423980	12/2018	King et al.	N/A	G06Q 30/0255
10424147	12/2018	MacKay et al.	N/A	G06K 7/00
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10741064	12/2019	Schwarz et al.	N/A	G08G 1/017
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11074612	12/2020	King et al.	N/A	G06Q 30/02
D927998	12/2020	Ahn et al.	D10/70	N/A
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D959298	12/2021	Jones et al.	D10/42	N/A
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11436649	12/2021	King et al.	N/A	G06Q 30/02
11462109	12/2021	King et al.	N/A	G06Q 30/00
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11699321	12/2022	Mackay et al.	N/A	N/A
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11776022	12/2022	King et al.	N/A	G06Q 30/0284
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11978300	12/2023	MacKay et al.	N/A	G07F 17/248
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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 15/750,273, filed Feb. 5, 2018, which claims priority to and the benefit of PCT Application No. PCT/CA2016/050928, filed on Aug. 8, 2016 by way of Canadian Application Nos. 2,900,177, filed Aug. 11, 2015, 2,908,793, filed Oct. 16, 2015, and 2,932,667, filed Jun. 9, 2016.

TECHNICAL FIELD

(1) The current application relates to parking meters and in particular to lightweight and vandal resistant parking meters.

BACKGROUND

(2) Traditional single space parking meters comprise a number of components. The metering functionality is typically provided by a meter mechanism that is received within a protective lower housing that is permanently fixed in location. A parking meter housing may be secured to the lower housing to retain the meter mechanism within the protective housing. These protective housing components are traditionally made either of a lighter weight cast metal such as Zamak which is an alloy made of zinc, aluminum, magnesium and copper, or of a much heavier cast ductile iron. Impact resistant plastic has previously been used as a housing material; however, this was typically limited to, for example, a polycarbonate resin. A specific example would be clear plastic windows made of a polycarbonate resin that covers and protects a parking meter display. A well-known clear polycarbonate brand in this regard is Lexan™. However, a Lexan window of a parking meter would not have been the material substantially or significantly forming the protective housing of a parking meter. As noted above, housing components have traditionally been made of metal.

(3) In the past, as parking costs and, correspondingly, primarily coins being deposited at parking meters have increased, vandalism and other attacks on parking meters have increased, and the use of metal housings has become standard for unattended (e.g., on street) locations in order to withstand and deter vandalism, including repeated attacks. In this regard, because of security concerns, plastic has historically been seen to be inferior to metal for the purpose of a protective housing of a parking meter. Further, plastic has been seen to be prone to undesirable UV degradation.

(4) As a result, at present, those skilled in the art generally prefer metallic parking meter housings given such housings are widely accepted and in use based on a strong track record for security and vandal-resistance, given use of plastics in parking meters has been overwhelmingly limited to internal molded plastic components and limited external components such as clear plastics for windows that cover a user display or solar panel, or plastics used to protect radiating antennas, and/or given the disadvantages of traditional polycarbonate or other traditional plastic material that would or may be felt if such material

were used in or for a parking meter housing.

(5) More recently, the use of cash as a form of payment for parking meters has been augmented by a number of alternative forms or methods of non-cash payment, which may include stored value smart cards, pay-by-cell mobile phone applications, loyalty cards, contactless credit cards, NFC-enabled mobile phones, magnetic stripe-based debit or credit cards, and EMV chip credit cards. With non-cash payment alternatives becoming increasingly popular, fewer and fewer users are carrying cash on their person or in their vehicles. It is not uncommon for the total revenue amounts and number of transactions based on these alternative non-cash payment methods to quickly surpass cash deposit amounts and cash transaction counts when such alternatives to cash payments are available.

(6) Although traditional metal parking meter housings provide adequate protection for parking meters, it would be beneficial to have additional or alternative options for parking meter housings capable of withstanding or resisting vandalism attacks.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments are described herein with reference to the appended drawings, in which:

(2) FIG. 1 depicts a single space parking meter;

(3) FIG. 2 depicts a further single space parking meter;

(4) FIG. 3 depicts a further single space parking meter;

(5) FIG. 4 depicts a rear cover of the parking meter of FIG. 3;

(6) FIG. 5 depicts an internal view of a front housing of the parking meter of FIG. 3;

(7) FIG. 6 depicts a further single space parking meter;

(8) FIG. 7 depicts the single space parking meter of FIG. 6 in a partially opened position;

(9) FIG. 8 depicts a multi space parking meter;

(10) FIG. 9 depicts a front isometric view of an alternative single space parking meter;

(11) FIG. 10 depicts a rear isometric view of the single space parking meter of FIG. 9 with the parking meter assembly partially raised;

(12) FIG. 11 depicts an internal isometric view of a front cover of the single space parking meter of FIG. 9;

(13) FIG. 12 depicts a rear isometric view of an internal structure of the single space parking meter of FIG. 9;

(14) FIG. 13 depicts a side view of the internal structure of FIG. 12 attached to the front cover of FIG. 11;

(15) FIG. 14 depicts a top perspective view of the front cover and removable components of the parking meter of FIG. 9;

(16) FIG. 15 depicts a detail perspective view of a removable contactless payment reader;

(17) FIG. 16 depicts details of the mounting of the removable contactless payment reader of FIG. 15 to the internal structure of FIG. 12;

(18) FIG. 17 depicts a side view of the front cover, internal structure and solar panel assembly;

(19) FIG. 18 depicts a side view of the front cover, internal structure, solar panel assembly and radio assembly;

(20) FIG. 19 depicts a top view of the radio assembly;

(21) FIG. 20 depicts a rear perspective view of the front cover, internal structure and locking assembly;

(22) FIG. 21 depicts a perspective view of the rear cover, base plate and locking assembly; and

(23) FIG. 22 depicts details of service locks.

DETAILED DESCRIPTION

(24) In accordance with the present disclosure there is provided a parking meter comprising: parking meter components for metering use of one or more parking spots; and, a housing formed at least significantly of a material comprising one of: a polycarbonate copolymer or polycarbonate blended resin; a nylon-based copolymer; or, a nylon-based polymer that includes long glass fiber or Kevlar fiber; wherein the housing encloses the parking meter components to provide protection for the parking meter components against an external environment and vandalism attacks.

(25) In a further embodiment of the parking meter, the housing comprises a plurality of housing components secured together, each of the housing components formed substantially of the housing material.

(26) In a further embodiment of the parking meter, the housing components comprise a front housing component and a rear housing component, wherein the front and rear housing components are securable together.

(27) In a further embodiment of the parking meter, the housing components comprise an upper housing component and a lower housing component, wherein the upper and lower housing components are securable together.

(28) In a further embodiment of the parking meter, wherein the housing further comprises one or more reinforcing members secured to or that are part of the housing.

(29) In a further embodiment of the parking meter, the one or more reinforcing members are secured to an interior of the housing.

(30) In a further embodiment of the parking meter, the one or more reinforcing members are secured to the housing by one or more mechanical fasteners.

(31) In a further embodiment of the parking meter, the one or more reinforcing members are secured to the housing through one or more formed structures of the housing mechanically retaining the one or more reinforcing members.

(32) In a further embodiment of the parking meter, the one or more reinforcing members are secured to the housing through one or more of a friction fit, interference fit or snap-in fit.

(33) In a further embodiment of the parking meter, the one or more reinforcing members are secured to the housing through an over-molding process.

- (34) In a further embodiment of the parking meter, the one or more reinforcing members are formed from a material comprising at least one of: a metal or metal alloy; and, a ceramic.
- (35) In a further embodiment of the parking meter, the one or more reinforcing members comprise one or more metal rods to provide protection against cutting type vandalism attacks.
- (36) In a further embodiment of the parking meter, the one or more reinforcing members reinforce one or more connection points of the housing.
- (37) In a further embodiment of the parking meter, the one or more reinforcing members comprise one or more metal structures at least partially surrounding one or more openings within the housing.
- (38) In a further embodiment of the parking meter, the one or more metal rods are secured to the housing through retaining structures or vertical channels of the housing.
- (39) In a further embodiment of the parking meter, the retaining structures or vertical channels extend vertically along a substantial portion of an interior surface area of the housing.
- (40) In a further embodiment of the parking meter, the one or more reinforcing members comprise at least one of: a wedge; a truss; a rib; a cross-brace; and, a stringer.
- (41) In a further embodiment of the parking meter, the housing material includes a mixture of small glass or ceramic beads or microspheres.
- (42) In a further embodiment of the parking meter, the housing material comprises one of a polycarbonate copolymer, a polycarbonate blended resin, or a nylon-based copolymer, and the housing material includes long glass fiber (LGF) or Kevlar fiber.
- (43) In a further embodiment of the parking meter, the housing material provides low attenuation of RF signals.
- (44) In a further embodiment of the parking meter, the parking meter components comprise one or more RF transmitters.
- (45) In a further embodiment of the parking meter, the one or more RF transmitters comprise one or more of: a near-field communication (NFC) reader; a Bluetooth radio; a ZigBee radio; a WiFi radio; and a cellular radio.
- (46) In a further embodiment of the parking meter, the one or more RF transmitters comprises a WiFi radio, and the WiFi radio uses one or more service flow channels associated with one or more WiFi routers located within wireless communication range of the parking meter.
- (47) In a further embodiment of the parking meter, the housing comprises one or more openings through which one or more buttons or interaction control components protrude, extend or are exposed.
- (48) In a further embodiment of the parking meter, the housing comprises an opening through which a display is visible.
- (49) In a further embodiment of the parking meter, the housing comprises a transparent portion through which a display is visible.
- (50) In a further embodiment of the parking meter, the housing comprises an opening through which a solar panel is exposed, or a recess in which a solar panel is received.
- (51) In a further embodiment of the parking meter, the parking meter comprises a transparent material through which a solar panel is exposed.
- (52) In a further embodiment of the parking meter, the parking meter comprises a contactless payment reader that is mounted: behind a surface of the housing; or, on top of a surface of the housing and is covered by a protective covering.
- (53) In a further embodiment of the parking meter, the parking meter comprises two or more flip-dots, and the housing comprises an opening or window through which the flip-dots are visible.
- (54) In a further embodiment of the parking meter, the parking meter further comprises input controls, and wherein the display and input controls are provided by a touch screen display.
- (55) In a further embodiment of the parking meter, the parking meter comprises a touch or proximity sensor located on the housing for detecting an initial user proximity or interaction with the parking meter.
- (56) In a further embodiment of the parking meter, the parking meter is a single space parking meter.
- (57) In a further embodiment of the parking meter, the housing comprises a first portion secured to a stationary structure and a second portion that is lockably secured to the first portion, at least some of the parking meter components secured to the second portion of the housing.
- (58) In a further embodiment of the parking meter, the housing is generally cylindrical in shape.
- (59) In a further embodiment of the parking meter, the second portion of the housing slides vertically relative to the first portion of the housing into a secured position.
- (60) In a further embodiment of the parking meter, the first and second portions of the housing comprise one or more rails or rail guides, respectively, and at least one of the one or more rails or rail guides comprises one or more of: a spring-catch assembly; and, a removable or detachable rail or rail guide for allowing release, separation or detachment of the second portion relative to the first portion.
- (61) In a further embodiment of the parking meter, the parking meter further comprises a hinge joining the first and second portions and which allows rotation of the second portion relative to the first portion.
- (62) In a further embodiment of the parking meter, the parking meter is a multi-space parking meter.
- (63) In a further embodiment of the parking meter, the housing material comprises a polycarbonate copolymer or polycarbonate blended resin.
- (64) In a further embodiment of the parking meter, the housing material comprises a nylon-based copolymer.
- (65) In a further embodiment of the parking meter, the housing material comprises a nylon-based polymer that includes long glass fiber or Kevlar fiber.
- (66) In accordance with the present disclosure there is provided a parking meter comprising: parking meter components for

metering use of one or more parking spots; a contactless payment reader; and, a housing formed at least significantly of a material comprising one of: a polycarbonate copolymer or polycarbonate blended resin; a nylon-based copolymer; or, a nylon-based polymer that includes long glass fiber or Kevlar fiber; wherein the housing encloses the parking meter components to provide protection for the parking meter components against an external environment and vandalism attacks.

(67) In a further embodiment of the parking meter, the contactless payment reader is mounted: behind a surface of the housing; or, on top of a surface of the housing and is covered by a protective covering.

(68) In accordance with the present disclosure there is provided a parking meter comprising parking meter components for metering use of one or more parking spots; a low-power short-range radio frequency (RF) radio for receiving payment information from a mobile device or vehicle; and a housing formed at least significantly of a material comprising one of: a polycarbonate copolymer or polycarbonate blended resin; a nylon-based copolymer; or a nylon-based polymer that includes long glass fiber or Kevlar fiber; wherein the housing encloses the parking meter components to provide protection for the parking meter components against an external environment and vandalism attacks.

(69) In a further embodiment of the parking meter, the low-power short-range RF radio comprises a Bluetooth radio.

(70) In a further embodiment of the parking meter, the Bluetooth radio is paired to the mobile device or vehicle prior to receiving the payment information.

(71) In a further embodiment of the parking meter, the received parking information comprises one or more of: a payment amount; a park time; an expiry time; and payment authorization details.

(72) In a further embodiment of the parking meter, the RF radio is further for communicating meter diagnostics information.

(73) In a further embodiment of the parking meter, the RF radio is further for communicating collected meter data.

(74) In a further embodiment of the parking meter, the RF radio is further for configuring the parking meter.

(75) In a further embodiment of the parking meter, the low-power short-range RF radio is at least partially powered by an alternative energy harvesting mechanism or device that harvests energy from one or more of: ambient RF signals; variations or differentials in nearby materials temperatures; and, street vibrations.

(76) Plastic components have been used with success in single space parking meters for many years. These include some components attached to the housings and exposed to the external elements. Specific component examples include clear plastics such as Lexan for the domes or windows that cover a user display, LED indicators, RF antennas, or a solar panel. Additionally, internally to the parking meter, components made of molded plastic are generally widely accepted and used. Specific component examples may include coin guides, and coin reader covers and chute. However, until now the exterior housings that protect the meter mechanisms have traditionally been made of cast metal alloy or cast ductile iron. The heavy duty appearance of these cast metal housings discouraged vandalism attacks and were sufficiently strong to withstand repeated vandalism attempts. As described further below, the heavy and expensive to manufacture metal housings can be replaced with a lightweight and vandal resistant housing made primarily of a polycarbonate copolymer material such as a polycarbonate copolymer resin that has lower corresponding materials and manufacturing costs. In this regard, a majority, a substantial majority or virtually all of the external parking meter housing may be made of a polycarbonate copolymer material.

(77) A lightweight housing, including preferably a plastic housing such as a polycarbonate copolymer resin housing, wherein the housing is formed substantially, or at least significantly, of the polycarbonate copolymer, may be easier and cheaper to transport and install, and may require less packaging, as compared to heavier metal housings. A housing formed at least “significantly” of the polycarbonate copolymer is one where at least one third of the housing is formed of the polycarbonate copolymer. A housing formed “substantially” of the polycarbonate copolymer is one where at least three quarters of the housing is formed of the polycarbonate copolymer. In addition, the housing may comprise a plurality of housing components secured together. Individually, each of the housing components may be formed substantially of the polycarbonate copolymer. Alternatively, a subset of the housing components may be formed substantially of the polycarbonate copolymer while the remainder of the housing components are not formed substantially of the polycarbonate copolymer, but the housing components taken together collectively provide for a complete housing that is formed significantly of the polycarbonate copolymer.

(78) A plastic housing, including a polycarbonate copolymer resin housing, may also be cheaper to manufacture than a metal housing, and, due to the lighter weight, cheaper to transport and easier for service staff to handle. Additionally, metal housings may be more inconvenient and/or costly than plastic housings, including a polycarbonate copolymer resin housing, in relation to maintenance. In particular, metal housings may require regular or periodic paint in order to avoid rust buildup or corrosion of housings. Sometimes, due to rust or corrosion, a metal housing may need to be replaced. Furthermore, rusted or corroded metal housings may be unsightly for the general public, thus reducing public confidence, and may detract from the visibility of newer or retrofit features otherwise included in the parking meter. In contrast, plastic housings, including a polycarbonate copolymer resin housing, are corrosion-resistant.

(79) In addition to the maintenance issues discussed above, metal housings, such as traditional sand cast steel housings, can produce imprecise joints and/or surfaces, which typically require secondary machining operations to allow any mating parts to fit properly. While Zamak alloy castings have a relatively more precise fit as compared to sand cast housings, both Zamak alloy castings and sand cast housings still require treatment and painting before being exposed to the outside environment. Secondary operations may also be required to remove metal ‘flash’ or burrs that might form at joints where mold sections are connected during the molding operation. In contrast, molded plastic-based materials typically provide for even more precise tolerances and dimensional stability that are better than those of traditional cast metal parts. This ultimately may produce smaller gaps at joints, and a better and more secure fit. Further, plastic-based materials, including polycarbonate copolymer resins, may not require the above-described secondary operations, such as painting or blasting. In this regard, plastics can be

produced in different colours and may not require the painting typically required for metal parts.

(80) Metal components may break or deform under sudden and focused impact. In contrast, polycarbonate copolymer resin-based plastics may temporarily bend or deform under impact and subsequently may return to their original shape in a manner that is similar to how a plastic car panel or bumper may do after an impact. Even at extreme cold temperatures, a polycarbonate copolymer resin will resist shattering or breaking. Unlike traditional (non-copolymer) polycarbonate, which may require considerable cross-ribbing or bracing material in order to achieve acceptable impact testing results or standards, a polycarbonate copolymer resin may not require as much reinforcing material, or any reinforcing material at all, in order to achieve desired results or standards (depending on the results or standards). Given its favorable strength and impact characteristics, and the decreasing prevalence of actual cash stored in parking meters (due to the increased use of non-cash payment means such as credit cards), a plastic housing such as a polycarbonate copolymer resin-based housing is a suitable alternative material to more traditional metals.

(81) Production of more complex shapes may more easily be done with plastic than metal due to injection molds, which allow for under-cuts, threads, ports and tight tolerances. This allows for a net shape or near net shape to be produced relative to the finish level specifications, which reduces the need for any surface finishing (i.e., a part may be produced near finish or fully finished without extra production steps, thus potentially saving costs).

(82) Plastic housings and parts, including polycarbonate copolymer resins, may be advantageous in comparison to metal housings and parts for additional reasons, such as with respect to the following: by using lower temperatures, plastic materials may be highly repeatable in processing, which may ultimately result in less scrap materials or waste; and the life of a plastics production tool may be as much as six times longer than the life of an equivalent tool used to produce a metal part, and may require less repair.

(83) Although the following description refers to single space parking meters, it is possible for the parking meters to provide metering functionality for two or more spaces. It will be appreciated that a single space parking meter refers to a style of parking meters that historically metered only a single parking space. Single space parking meters are typically located along a street and affixed to posts adjacent to the parking spaces being metered. As an example of metering multiple parking spots, a single space parking meter located on a post in between two parking spots may be configured to provide metering of the parking space located to the left of the parking meter as well as the parking space located to the right of the parking meter. Single space parking meters are typically differentiated from multi-space meters, such as pay-and-display meters, pay-by-space meters and pay-by-plate meters, by their size. Multi space meters, which are typically provided in larger cabinets, are typically located one or a few per block, or per parking area such as a parking lot. The larger cabinets of multi space meters may allow for additional peripheral equipment such as thermal type receipt or ticket printers, paper currency readers, full alpha-numeric keypads, higher capacity cash cans, and correspondingly larger batteries and solar panels. In contrast, multiple single space meters, even when metering multiple parking spots, are typically located along a block face, or within a parking area such as a small parking lot. The single space meters are typically much smaller in size than multi-space meters.

(84) FIG. 1 depicts a typical single space parking meter **100**. The parking meter **100** comprises a coin vault housing **102** secured to a pole **104** or other structure at a parking meter location, such as an on-street location. The coin vault housing **102** holds a coin vault that holds coins used for payment. A lower housing **106** may be secured to the coin vault housing **102**. The lower housing **106** receives a parking meter mechanism that provides metering functionality. An upper housing **108** may be secured to the lower housing **106** in order to secure the parking meter mechanism within the lower housing **106**. The lower housing **106** may be permanently or semi-permanently secured to the coin vault housing **102**. The meter mechanism may be replaced with a retrofit parking meter mechanism that is received within the same lower housing. The retrofit parking meter mechanism may be secured to the lower housing **106** using the same upper housing **108**, or may be secured using a different upper housing. The upper housing of the retrofit parking meter may be made from a polycarbonate copolymer resin providing strength and protection against vandalism as well as the environment. The polycarbonate copolymer resin housing may be a separate component or may be incorporated into the retrofit parking meter mechanism itself. As described in further detail below, the polycarbonate copolymer resin housing of the parking meter may include additional reinforcing components, or structural ribbing detail molded into the plastic material, in order to provide additional mechanical strength.

(85) FIG. 2 depicts a disassembled view of a further single space parking meter **200**. The parking meter **200** comprises a lower housing **202** that may be permanently, or semi-permanently, installed at a parking meter location. A removable parking meter mechanism **204** may be partially received within the lower housing **202**. The removable parking meter mechanism **204** may have an internal frame **206** to which parking meter components are attached. The parking meter components may include, for example, payment mechanisms, including coin chutes, card readers, and contactless card readers. The parking meter components may also include buttons **208** or other interaction control components for controlling operation of the parking meter, as well as a display **210** for displaying parking related information such as an amount of time purchased. Further, the components may include one or more radio frequency (RF) antennas for transmitting and/or receiving RF signals. In this regard, such RF transmitters may comprise one or more of a near-field communication (NFC) reader, a Bluetooth radio, a ZigBee® radio, a WiFi® radio, and a cellular radio.

(86) With respect to the WiFi radio, it may use a private communication channel, or any public or otherwise available hotspot for carrying out communications. For example, an owner or operator of one or more parking meters may engage or contract with an Internet service provider (ISP) to provide for Internet access over WiFi for the parking meter(s). In one example, the arrangement with the ISP may involve the use of additional or secondary channels, such as secondary ISDN channels that may be referred to as “service flow” channels, associated with one or more WiFi routers located within wireless communication range of the parking meter(s). Typically, the WiFi routers in this example are located at an ISP's customer premises, and are such that they have the ability to concurrently route or support one or two additional or secondary non-

overlapping and non-interfering ISDN service flow channel(s). The service flow channel(s) is in addition to the primary ISDN channel used by the customer, and all channels use the same wire or other physical connection to the ISP's network equipment located outside the customer premises. The result is that the router acts as a secured WiFi hotspot for one or both of the two additional channels, and, in parallel, routes any associated data or traffic separately and independently from the data and traffic of the primary ISP customer. Use of one or more of the additional service flow channels by a parking meter system operator establishes a connection between a parking meter and the Internet so that regular two-way flow of data between the meter and a backend meter management system is possible.

(87) The foregoing describes one example of how a WiFi radio and communication system may be used; other approaches would be understood to be possible. Use of WiFi as described or otherwise for parking meter communications may be advantageous, as purchasing or inclusion of additional hardware by parking meter vendors, installers or operators may be unnecessary. This is in contrast to cellular communications and other alternative ISM band solutions such as ZigBee where additional hardware may be required. In this regard, and by way of example, a transceiver or repeater with antenna may need to be located relatively close to a parking meter, and may need to be mounted to a pole, building or other structure, which may further complicate installation and/or implementation of a parking meter system. In addition, use of WiFi as described or otherwise may be advantageous in that regulatory requirements (e.g., FCC requirements) may be less restrictive or onerous, the WiFi radios may be less expensive than other radios, and the data transmission costs associated with cellular communications may be reduced or eliminated as a result of a greater amount of data being transmitted over WiFi as opposed to cellular. Further, with respect to the example approach described above, reliability of transmissions may be improved because of the use of a managed, secured and/or restricted Internet access point (as compared to a public WiFi hotspot, which may be less reliable and/or provide poor performance due to heavy use or data loading).

(88) Since the housing is made from a polycarbonate copolymer resin that can have low RF attenuation, the placement of the RF antennas are not as particularly restricted as is the case for more traditional metal housings. The removable parking meter mechanism **204** includes a front housing **212**. The front housing **212** is made of a polycarbonate copolymer resin, and includes one or more cutouts or openings through which interface buttons **208** extend, protrude or are exposed, and through which the display is visible. The display, when visible through a cutout in the polycarbonate copolymer resin housing, may be covered by a clear polycarbonate material such as Lexan. In the alternative, rather than a physical opening or cutout in the polycarbonate copolymer resin, the display opening may be provided by a transparent portion in the housing that may be formed from the transparent polycarbonate copolymer resin of the housing **212**.

(89) Physical openings in the polycarbonate copolymer front housing **212** may present a weak point in the housing **212** and as such the housing **212** may include reinforcing members to provide it with additional strength. For example, a reinforcing member may comprise a metal frame or structure surrounding the opening and be located on an interior of the front housing **212**. The reinforcing member surrounding the display opening may provide additional strength to prevent a vandal from prying at the opening, or otherwise attacking the integrity of the housing through the opening. As depicted, when the removable parking meter mechanism is received within the lower housing, a rear housing **214** may be slid over the removable parking meter mechanism **204** and secured to the lower housing. When secured in place, the front housing **212** and rear cover **214** provide a continuous housing providing protection against the environment and vandal attacks. In addition to providing reinforcing members at, or around, openings, metal reinforcing members may also be provided for structurally weaker areas of the housing such as at sharp corners. In addition or in the alternative to metal reinforcing members, the polycarbonate copolymer component may include, as part of its design and molded shape, wedges, trusses, ribbing, cross bracing, stringers, and the like that are more precisely and strategically shaped and placed, and are designed into the injection molded component for reinforcement. This may further enhance the structural strength and integrity of the component without adding substantially more or additional plastic material. As a result, the polycarbonate copolymer resin-based component that is produced may provide further improved strength based on the combination of above-described mechanical design features together with the inherent strength and ductility characteristics of the polycarbonate copolymer resin materials.

(90) The rear cover **214** may be made from the same polycarbonate copolymer resin used to make the front cover **212**. As described above, the polycarbonate copolymer housing may include at openings reinforcing components to provide extra strength. As depicted, the rear cover **214** includes an opening with reinforcing member **216** surrounding the opening. Further, reinforcing members or components may be included to provide extra strength to connection points. For example, the rear cover **214** may include four contact points **218a-218d** used to secure the rear cover **214** to the lower housing **202**. In order to provide additional strength, the contact points **218a-218d** may be reinforced with metal. The reinforcing members may be formed within, or secured to, the polycarbonate copolymer, for example by an over-molding process in which the polycarbonate copolymer resin is formed over the metal reinforcing contact points. Alternatively, the reinforcing members may be secured to the polycarbonate copolymer housing in various ways such as by a mechanical fastener, such as screws, through formed structures that co-operate with the reinforcing member, through a friction fit, interference fit or snap-in fit and/or through the use of adhesives.

(91) As described above, the single space parking meter **200** comprises a housing made from a polycarbonate copolymer resin that may be reinforced at one or more points with metal strengthening components. As depicted, the housing may be made from two or more housing components that connect together to provide a single housing that encloses the parking meter components and protects them from the environment and from vandal attacks. As depicted, the housing may secure the removable parking meter mechanism to the lower housing **202**. As depicted in further embodiments below, it is possible for removable parking meter mechanisms to be directly secured to the coin vault rather than to a lower housing.

(92) FIG. 3 depicts a further single space parking meter. The parking meter **300** comprises a removable parking meter assembly **302** that is connected to a coin vault assembly **304**, rather than being received within a traditional single space

parking meter lower housing. The coin vault housing provides an opening, secured by a door or covering (not shown), through which a coin vault or canister can be inserted and removed. The coin vault assembly may include or be attached to a lower protrusion **306** for use in directly securing the coin vault assembly, and indirectly the attached parking meter assembly, to a post or other structure.

(93) The parking meter assembly **302** comprises a front covering **308** formed from a polycarbonate copolymer resin. The parking meter assembly **302** further includes a number of parking meter components for providing a functioning parking meter. Particular components included in a parking meter may vary depending upon the manufacturer, or the specific requirements of a city, municipality or other potential purchaser of parking meters.

(94) The parking meter assembly **302** may be secured to a rear housing assembly **324** that in turn is secured to the coin vault housing **304**. As depicted, the rear housing assembly **324** comprises a rear cover **326** that is affixed to a lower plate **328**. The lower plate **328** may be permanently, or semi-permanently, attached to the coin vault housing. The rear cover **326** may be secured to the lower plate through one or more connecting bolts or rods **330** within corresponding retaining structures or vertical channels in the rear cover **326** that may be screwed into the lower plate **328**. The bolts or rods **330** are of a hardened material such as metal, and may be threaded. The rear cover **326** may be formed from the polycarbonate copolymer resin used in forming the front cover **308**.

(95) In addition to securing the rear cover **326** to the lower plate **328**, the connecting rods **330** may act as reinforcing members that provide additional strength and protection against vandalism. For example, the rods **330**, which may be made of metal, may extend vertically along a substantial portion of the interior perimeter or surface area of the rear cover **326**. The metal rods **330** may provide additional protection from vandal attacks. In this regard, the rods may act as reinforcements and may be used to augment the high strength housing. As a result, the rods may act as a deterrent to vandal attack using hack saws, or other cutting tools. Similar reinforcements may be applied to injection molded covers or housings in strategic areas, such as connection points, surrounding openings, or in likely points of attack by vandals. The reinforcing metal components may be inserted into formed channels, pockets, or similar structures for receiving the reinforcing components. Additionally or alternatively, the reinforcing metal components may be incorporated into the injection molded covers using an over molding technique.

(96) FIG. 4 depicts an isometric view of the rear cover **326** in further detail. The rear cover **326** may be formed as an extrusion of a polycarbonate copolymer resin that is cut to the required length and subsequently machined to provide the desired component. Alternatively, the rear cover **326** may be formed from injection molding of the polycarbonate copolymer resin. The rear cover **326** may include one or more molded or formed rails **402a**, **402b** on which the front cover **308** of the parking meter assembly **302** may be captured and slide up and down. Further, the interior of the rear cover comprises a plurality of extended cylindrical portions, vertical channels or retaining structures **404a**, **404b**, **404c**, **404d** that extend down a length of the rear cover and through which the connecting rods may be inserted. The connecting rods may extend through the bottom plate **328** and the extended cylindrical portions **404a-d**. A nut may be used to secure the bolt or rod, and so the rear cover **326** to the bottom plate **328**. Alternatively, the rear cover **326** and the bottom plate **328** may be formed as a single injected component.

(97) The rear cover may further include one or more openings **406** for elements such as LEDs or other visual devices for quickly communicating a status of the parking meter to parking officials. Further, the rear cover **326** may include an opening **408** for a locking component. As with the front cover **308**, the openings **406**, **408** of the rear cover may also be surrounded by a reinforcing metal component to harden or strengthen the opening in the rear housing for protection against attacks.

(98) The front cover **308** and rear cover **326**, which are shown in FIG. 3, may be securely joined together forming a secure housing to protect the parking meter components against an external environment, including dirt, dust, rain and snow, as well as from vandalism. Although the secure housing is depicted as being provided by two interacting housing components, namely, the front cover **308** and the rear cover **326**, it is contemplated that additional covers, or portions of the housing, may be provided. When in use, the rear cover **326**, front cover **308**, parking meter components and bottom plate are coupled together and secured to the coin vault housing **304** by a locking mechanism. The locking mechanism may be provided by a single locking component securing all of the parking meter assembly components together. Alternatively, the locking mechanism may include a plurality of locking components.

(99) The rear cover **326** and the front cover **308** of the parking meter **300** may provide a generally cylindrical appearance or shape. Although the appearance of the parking meter **300** may be a design aesthetic that may be varied based on different design factors, a cylindrical appearance may provide a more attractive parking meter assembly than a non-cylindrical assembly. In addition to providing a visually more attractive appearance, the cylindrical shape may improve the strength of the parking meter for withstanding physical attacks since the cylindrical nature of the housing eliminates, or at least reduces, the number of corners in the housing, which corners may provide weaker points. The front cover **308** may have a recessed flat portion that provides a location **310** for user interaction. The user interaction location **310** may comprise an information area or display **312** for presenting information to a user, as well as input controls **314** for interacting with and controlling the parking meter assembly **302**.

(100) The recessed flat portion may also include a payment area **316** for providing a plurality of different payment mechanisms. The plurality of different payment mechanisms may be arranged, one next to the other, horizontally. For example, a coin chute, including an opening **318** for inserting coins into the coin chute, may be provided along with a card slot **320** for accepting payment via a magnetic stripe or chip card. The card slot **320** as depicted is vertically-oriented. Optionally, the card slot may be horizontally-orientated and/or angled so as to shed away any water in the vicinity of the card slot (e.g., the card slot may be sloped downwardly away from the interior of the parking meter). The payment area **316** may also have a tap area **322** for making payment via a contactless payment device. The tap area **322** may include an industry-

recognized logo and/or a message such as “tap here” for directing a user to where the contactless payment device should be held to effect payment. The tap area **322** may also include one or more progress indicators, which may for example be a band of light emitting diodes (LEDs) for providing an indication of the progress of a transaction. The progress indicators may include color-based indications (e.g., green indicates a successful transaction).

(101) Although depicted as a separate tap area **322** in the payment area **316**, contactless payment readers and/or antennas therefor may be provided at other locations including, for example, surrounding, or in close proximity to, the display **312**, or on a lower non-vertical or angled surface that provides a transition from the recessed flat portion to the cylindrical perimeter of the front cover **308**. The contactless payment readers and/or antennas therefor may alternatively be mounted such that they are attached to or mounted on top of a surface of the housing, such as on top of one or more surfaces of the front cover as opposed to being flush mounted and/or hidden behind a front cover surface(s). In this regard, a number of locations for the contactless payment reader and corresponding antenna are possible. For example: a self-contained contactless payment reader device, such as an after-market contactless payment reader device may include a built-in antenna, and may be located or mounted either behind the front cover surface or on top of the front cover surface; or, alternatively, a contactless payment reader device may be physically separate from but connected to its corresponding antenna, and either one or both of the reader and the antenna may be located or mounted either behind the front cover surface or on top of the front cover surface (such that if the antenna is mounted externally and the reader is mounted internally, there would be a connection between the two through the front cover). For instances where the contactless payment reader and/or antenna are mounted on the surface of the front cover, they may be mounted by way of bolt and/or adhesive, and the may be protected by suitable plastic or other protective covering for protection against vandalism and the elements. In the case of a contactless reader mounted on the surface of the front cover, it may extend outwardly from such surface or sit higher on the surface. The front cover **308** may be made from a polycarbonate copolymer resin, which may be substantially transparent to radio frequency (RF) signals, allowing greater flexibility in where the tap area for the contactless payment reader may be located. In addition, and depending on the layout of components, either the user interaction area **310** or payment area **316** may include a keypad (not shown) for entering information, such as a PIN or license plate number for carrying out transactions. Further, the recessed flat portion that provides a location for user interaction may be illuminated with one or more lights (not shown), such as LED-type lights, that may be positioned at or near where the top of the flat portion transitions from the recessed flat portion to the cylindrical perimeter of the front cover **308**.

(102) The parking meter may include a low power and short range radio frequency (RF) type radio. The low power RF radio may be a Bluetooth® type radio, although other types of RF radios are possible, including, for example, WiFi, ZigBee, etc. More specifically, the low power and short range RF radio may be a Bluetooth low energy (also known as Bluetooth LE, BLE or Bluetooth Smart) radio, which provides lower power consumption than traditional Bluetooth but maintains a similar communication range. The RF radio may be mounted within the interior of the housing and may be constantly supplied with power, or supplied with power while there is sufficient power to power the RF radio. The housing, which may be made either significantly or substantially of polycarbonate copolymer resin, will not significantly attenuate RF signals from the RF radio. Accordingly, the housing may provide flexibility in the placement and mounting of the RF radio within the meter interior. The RF radio may have an electrical connection, or interface, to the meter controller board allowing the constantly powered RF radio to notify or interrupt the controller board, even if the meter is in a low powered or sleep state, as well as send and receive data and/or commands to and from the controller. By having the RF radio continuously powered, there is no need for a user having a mobile device to initially interact with the meter such as by pressing a user button, inserting a card, etc. in order to ensure the parking meter is in a wake, or operation, mode. Rather, the user, or more particularly the user's mobile device communicating with the RF radio, may place the meter in the wake or operation mode.

(103) The primary power source for the Bluetooth low energy radio may comprise one or more battery packs located inside the parking meter. A supplementary source of power for the Bluetooth low energy radio, and/or for other devices, components or circuits within the parking meter, may comprise an alternative energy harvesting mechanism or device. For example, the parking meter may include an alternative energy harvesting mechanism or device that harvests energy from ambient RF signals, variations or differentials in nearby materials temperatures, and/or street vibrations. Energy harvested in this regard may be used to at least partially power the Bluetooth low energy radio and/or other devices, components or circuits within the parking meter. Such supplemental power source may allow a device such as a Bluetooth low energy radio to be more autonomous from and self-sufficient relative to the primary parking meter power sources and circuits, including the one or more battery packs of the parking meter.

(104) The RF radio device may be used to facilitate purchases of park time from mobile devices or vehicles which have paired, or otherwise associated, with it. An associated mobile app may be used by the owner of the mobile device or vehicle to request and make the park time purchase. An identifier or name of the RF radio device in a particular meter being paired with, or otherwise associated with, the mobile device or vehicle may be associated with the meter's location or an ID label found on the exterior of the meter housing. Such a visible identifier or name allows the mobile device or vehicle owner to visually check to ensure they are pairing with, or otherwise associating with, the desired meter from or within a group of nearby meters that may also be outfitted with RF radio devices and possibly be within reception range of the mobile device or vehicle. The application on the mobile device may also provide the ability to scan or read barcodes, 2-D ‘QR’ codes, or RFID tags attached to the exterior of the meter, which can provide additional information about the meter including the meter location identifier or ID. Scanning a bar code, QR code or RFID tag may reduce the amount of user keyboard entry, thus speeding up the park time request process. The owner of the mobile device or vehicle may have previously registered the mobile app and associated it with a payment means such as a credit or debit card or other financial payment means such as PayPal, Applepay, GoogleWallet or Android Pay.

(105) Once the mobile app or vehicle has paired with, or otherwise associated with, the selected RF radio device in a given meter, the current meter parking rate information is made available to the app, and the user completes their request for park time and confirms payment. Once payment is confirmed by the mobile app or vehicle, payment information, such as payment amount, park time, expiry time and/or authorization details may be provided from the app or vehicle to the meter via the RF radio device. The meter may then show paid status as well as the amount of park time purchased and/or remaining.

(106) The low power RF radio may also be used by parking authority meter repair staff (e.g., city or municipality staff) to interact with the meter using an app on their mobile devices, without the use of any other special service cards, pressing any buttons, or physically opening up the meter housing. The app may allow for meter diagnostics, meter data collection, meter configuration, as well as repair staff time management. This alternative meter communication channel is particularly helpful to parking authority meter repair staff if the meter's cellular radio is not functional or the meter is located in an area with particularly poor cellular coverage.

(107) The information area **310** may include one or more input controls **314** for interacting with and controlling the parking meter. For example, the input controls **314** may include buttons for increasing or decreasing an amount of time to be purchased, an 'OK' button, a 'cancel' button and one or more additional input controls which may be used for navigating one or more options or menus presented on a display **312**. For example, if the parking meter is for two parking spots, left and right buttons may be included so that a user may indicate which spot he/she is paying for. Individual buttons may alternatively be combined into a keypad array, arranged horizontally above or below the user display or vertically to the left or right of the display.

(108) The individual buttons and/or individual keys of a keypad array may use piezo-style switches, buttons or actuators, which are substantially or fully sealed from external environmental elements, thus making them advantageous for use in outdoor equipment. The keypad array may provide an electrical interface in the form of one or two conductors for each normally open or normally closed piezo-style switch, button or actuator in the array. Alternatively, to reduce the number of conductors, the keypad array electrical interface may use a circuit that implements a one, two or three-wire communications protocol, such as inter-integrated circuit (I2C) bus, serial peripheral interface (SPI) bus, or a custom protocol. In this regard, the number of input/output (I/O) lines or conductors required to detect switch states may be reduced or minimized. This form of electrical interface provides improved flexibility, as a keypad array with just a few buttons can use the same electrical interface as a keypad array with many buttons. For example a two-button, a six-button, or a 40-button keypad array could all be implemented using the same electrical interface. Further, reducing the number of conductors as described may also advantageously free up I/O on a meter's controller, and/or use an existing bus already used by the meter to manage other peripherals.

(109) In addition to the display **312** for presenting information to a user, the interaction area **310** may further include one or more status LEDs for communicating a status of the parking meter at a distance. For example, the status LEDs may be used by parking enforcement personnel to determine which parking meters have expired time, have purchased time remaining, are malfunctioning, etc. without having to closely inspect the meter. The status LEDs may employ one color or more than one color in order to convey one or more statuses to parking enforcement personnel with respect to the parking meter, which, for example, may be a parking meter for a single parking spot or two parking spots. It is possible to replace the status LEDs with other indications that can be understood from a distance, such as flashing of the display, or magnetic "flip dots". (While not depicted, the rear cover may also provide a clear window or opening that allows two or more brightly coloured and/or highly visible enforcement "flip-dots" (also known as "flip-discs"). The "flip-dots" may be magnetically energized, and each individual "flip-dot" may have a different colour on each side of it. A particular colour serves to represent a particular status of a specific parking space managed by the meter. Such "flip-dots" may be easily visible from enforcement vehicles or by walking parking attendants even in bright sunlight.) However, the status LEDs, particularly when flashing a low duty cycle, draw relatively very little power, and as such are preferable for remaining on for long periods of time, whereas powering the relatively larger display, even if only flashing, may consume a significantly greater amount of power. Additionally, although depicted as separate display and input components, it is contemplated that the input controls **314** and the display **312** may be provided by a single touch screen display, which may or may not provide color graphics. Although different types of touch screen sensors may be used, the preferred technology solution would offer low power consumption, which allows the touch screen sensor circuits to remain active even during non-operational periods, and when the main display is turned off to conserve power, so that any tap or touch to the display screen by the user will activate the meter. A touch screen display may allow some or all other user input controls to be eliminated. Alternatively, an additional touch or proximity sensor located anywhere on the meter front surface that preferably has low power consumption may be used to detect an initial user proximity or interaction with the parking meter, which may permit the parking meter to power on additional components such as the touch screen display. The preferred touch screen sensor would be robust, resistant to abuse and vandalism, and function reliably in outdoor environments, including extreme heat and cold, as well as with a gloved finger. Although not depicted in the FIGs, the parking meter may comprise a printer, including paper roll, and opening in the front cover for dispensing receipts for users upon payment of parking time.

(110) As is well understood, single space parking meters are powered by one or more internal battery packs. The batteries may be supplemented, and/or charged, by a solar panel **330**. Although other positions are possible, locating a solar panel **330** at or near the top of the parking meter is a desirable location. While the solar panel **330** is depicted in the FIGs as being in a fixed, forward sloping orientation, it will be appreciated that the cylindrical shape of the exterior housing allows for an adjustable solar panel assembly that can be rotated at the time of installation such that the solar panel may be oriented to point in any direction. Typically, the direction to be pointed is due south for the northern hemisphere. The adjustable solar panel assembly may be allowed to be adjusted to any one of two or more vertical slopes or angles relative to the horizon.

Although not depicted in the FIGs, the solar panel **330** at the top of the meter may also be optionally augmented with one or more additional flexible type solar panels attached to the exterior of the cylindrical surface of the rear cover.

(111) As depicted, the parking meter **300** may be comprised of two sub-assemblies, namely, a front assembly **302**, and a rear assembly **324**. The rear assembly **324** may be secured to a coin vault housing permanently secured to the parking meter location. The rear assembly **324** may comprise a bottom plate **328** secured to the rear cover **326**. The bottom plate can be permanently, or semi-permanently, attached to the coin vault housing. The front assembly **302** may be slidably engageable with the rear assembly **324** to allow the front assembly **302** to be removed from the rear assembly. The front assembly **302** may be fully removed, or partially removed or partially installed. That is, the front assembly **302** may engage with and slide down or up rails or profiles of the rear assembly **324** in order to secure the front assembly **302** to the rear assembly **324**. In this regard, the front assembly **302** may comprise rail guides that engage with the rails of the rear assembly **324**. The rail guides of the front assembly **302** may be formed as part of the front cover **308**, or may be provided as separate components that may be attached to the front cover **308**. It will be appreciated that the opposite configuration—i.e., rail guides on the rear assembly and rails on the front assembly—may also be possible, and that the variations and embodiments described herein may similarly be reverse vis-à-vis the front and rear assemblies. Additionally or alternatively, the front cover of the front assembly may include side edge surfaces that provide a bearing surface that in conjunction with bearing surfaces of an internal structure of the front assembly capture the rails of the rear assembly and allow the front assembly to slide up and down the rear assembly.

(112) The front assembly **302** may be locked to the rear assembly **324** using a locking mechanism, which may be operated via a keyhole in the rear cover. The locking mechanism (not shown) may be affixed to the rear cover and engage with and secure the front cover in the locked position, or alternatively the locking mechanism may be affixed to the front cover and engage with and secure the rear cover when in the locked position. The front assembly **302** may be removed and replaced as a complete assembly by fully raising or lifting the front assembly vertically until its bottom end clears the rails of the rear assembly. Partially or fully raising the front assembly **302** from the rear assembly **324** may be done to, for example, inspect components or perform routine maintenance such as replacing batteries or clearing card reader or coin chute blockages. The rails may provide a spring-catch assembly (not shown) that allows the front assembly **302** to be lifted and held at any one of multiple partially raised vertical positions which allows hands free service work and testing of the meter to be carried out while in the partially raised or lifted position. Releasing the spring-catch would release the front assembly **302** allowing it to freely slide up or down on the rails or profiles. Alternatively, the front assembly **302** may be raised to a first position such that the rail guides of the front assembly **302** may be released from the rails of the rear assembly **324** at this first position, while the bottom of the front assembly **302** remains engaged with the rear assembly. Under this alternative configuration, there would also be a hinge, such as a telescoping or folded metal rod, located at a release point. For example, the hinge may be fixed at or near the top of the front assembly at one of its ends (such as at or near the top of the rail guides) and at the other end fixed at or near the top of the rear assembly (such as at or near the top of the rails). Releasing, separating or detaching the front assembly from the rear assembly at the release point allows the hinge to unfold or extend to a stopping point and the front assembly to tilt forward to a non-vertical or horizontal position. The bottom of the front assembly that remains engaged with the bottom of the rear assembly is pivotably configured at such bottom location so as to allow the front assembly to rotate forward from its vertical position to the more horizontal position. Such pivotable configuration may require that the rail guides at the bottom of the front assembly be partially releasable or hinge-able relative to the bottom of the rails and/or bottom of the rear assembly in order to allow for the rotation of the front assembly. Such rotation, which involves pivoting or hinging about an axis at the bottom location, thus allows for the back side or internally-facing components of the front assembly to be exposed for service work. The ability to raise and/or hinge forward the front assembly in this regard is advantageous for more openly exposing some or all of the internal parking meter components as compared with traditional single space parking meters. This may make repair, servicing and/or replacement of such components easier and/or quicker. With respect to the coin chute, a mirror (not shown) may be positioned just above a coin inspection slot or opening for the purpose of facilitating or making easier the inspection of the coin chute for blockages. Alternatively, payment devices may have quick release knobs allowing them to be disengaged from a normally fixed/rigid position on the front cover, such that they can be quickly and easily inspected, serviced and/or removed without the use of tools.

(113) FIG. 5 depicts a rear isometric view of the front assembly's front cover or housing. The front cover is primarily formed from polycarbonate copolymer resins, which provides for lower interference than metal housings with respect to wireless communications from the parking meter. The front cover **308** may be molded or otherwise machined to provide the exterior surface, as well as to provide a plurality of mounting points **502** for connecting the parking meter components. The front cover **308** may include strategically placed features or projections on the inside surfaces for the attachment of steel rod or sheet metal reinforcing members which may provide additional vandal resistance against common cutting tools. The front cover **308** may comprise reinforcing members located around openings **504** in the housing. In addition or alternatively, the injected molded polycarbonate copolymer resin may include a mixture of small glass or ceramic beads or microspheres providing anti-drilling protection. Additionally, or alternatively, long glass fiber (LGF) or Kevlar® fiber may be included in the resin. Combining the raw plastic material with the above-described additional or alternative materials at the time of mold injection may result in a formed component that is particularly resistant to drilling by traditional carbide tip drills. The introduction of such additional or alternative materials may increase tensile strength and resistance to penetration, though such materials may correspondingly cause a reduction in elasticity, ductility and/or flexibility.

(114) The front cover **308** may include a recess **506** at the top of the front cover **308** for receiving a solar panel. The solar panel recess **506** may include a secondary radio antenna recess **508** located towards the front of the front cover **308**. The radio antenna recess **508** may provide a location for a flexible antenna that may be used for various communications,

including for example cellular data communications related to authorization of payment transactions, verification of parking meter status, and/or updating of parking meter functions. Preferably, the antenna is located at or close to the curved inner surface of the front cover **308** within the recess **508**. Alternatively, the antenna may be integrated into or surface mounted to the top facing surface of the solar panel. The solar panel recess **506** and radio antenna recess **508** may be fixedly part of, or rigidly attached to, the front cover **308**, or, alternatively, the solar panel recess **506** and radio antenna recess **508** may be removable components that are replaceable in the field. The solar panel may optionally be covered with a transparent material that may be applied after the solar panel is received in the solar panel recess. Also, the solar panel may be permanently sealed or glued in the recess using an adhesive, such as a 3M adhesive gasket. In the alternative, the solar panel may be provided in a solar panel assembly or module that slidably engages with or is removably coupled or secured to the front cover **308**, and further electrically engages with a battery of the parking meter when the solar panel assembly is fully engaged with the front cover. The solar panel assembly allows the solar panel to be replaceable/serviceable in the event of vandalism, failure or other damage. Such solar panel assembly may include the one or more lights, such as LED-type lights, that may be positioned at or near where the top of the flat portion transitions from the recessed flat portion to the cylindrical perimeter of the front cover **308**. Such solar panel assembly may further include status LEDs or holes for permitting the visibility of such status LEDs. In addition to the alternative solar panel assembly, the radio antenna may alternatively be provided in a radio antenna compartment that slidably engages with or is removably coupled or secured to the front cover **308**, and further electrically engages with the main electronics board when the radio antenna compartment is fully engaged with the front cover. This allows the radio and antenna combination to be quickly and easily swapped out in the field for maintenance/service, to replace/inspect a SIM card, or to switch the meter from one cellular network to another alternative network that may have better coverage or signal strength than the original network. The inside surface of the front cover **308** comprises a number of mounting locations for other components. The mounting points may include, for example, a mounting location for display and input components, a mounting location for a contactless reader, a mounting location for a magnetic stripe and chip card reader as well as a mounting location for a coin chute.

(115) Although a specific molding of a front cover is depicted, it will be appreciated that the particular locations of the mounting points and other features may vary, and will depend upon the specific components incorporated into the parking meter.

(116) FIG. **6** depicts a further single space parking meter and FIG. **7** depicts the single space parking meter of FIG. **6** in a partially opened position. The parking meter **600** comprises a removable meter mechanism **602** that is received within a lower housing **604**. The removable meter mechanism **602** comprises a front housing **606** formed of a polycarbonate copolymer resin. The front housing **606** comprises a number of openings including openings for buttons or control components **608** and a display **612**. Situated on top of and affixed to a flat recessed portion of housing **606** is a solar panel **610**. Alternatively, the solar panel **610** may be exposed to light through an opening in the front housing **606**. As described above, the openings within the front housing **604** may be strengthened by one or more reinforcing metal members. As depicted in FIG. **7**, the front housing **606** may be pivotally connected to an internal portion received within the lower housing **604**. A rear housing **614** may be secured to the lower housing **604**, and may also be formed of a polycarbonate copolymer resin. The front housing **606** and the rear housing **614** may be secured together in order to provide a unitary housing providing protection from vandalism attacks.

(117) FIG. **8** depicts a multi space parking meter **800**. The parking meters described above are single space parking meters. However, the lightweight polycarbonate copolymer housing may also be used in multi space parking meters. As depicted, the multi space parking meter **800** may have a housing **802** formed of a polycarbonate copolymer resin. Additional reinforcing members may be provided on an interior of the multi space parking meter housing in order to provide additional strength and protection against vandal attacks. For example, the parking meter housing may include metal rods at select locations to prevent sawing or cutting type attacks. Additional reinforcing members may be located surrounding openings for other components such as payment devices, displays, compartment doors, etc. In addition, the polycarbonate copolymer resin may include a mixture of small glass or ceramic beads or microspheres providing anti-drilling protection.

(118) FIG. **9** depicts a further parking meter. The parking meter **900** is similar to the parking meters described above. In particular, a parking meter assembly comprises a protective outer covering, which may be made at least in part from a polycarbonate copolymer resin that provides protection to internal parking meter components. The protective outer covering may be provided by a plurality of exterior covers. One or more of the covers may be formed substantially of a polycarbonate copolymer resin. Alternatively, it is possible for an individual cover to be formed from something other than a polycarbonate copolymer resin, or for an individual cover to be formed of a polycarbonate copolymer resin, but not substantially of a polycarbonate copolymer resin. Preferably, at least one cover is formed substantially of a polycarbonate copolymer resin. For example, the protective outer covering may be provided by a front cover that is formed substantially of a polycarbonate copolymer resin and a rear cover that is formed from metal. Alternatively, the rear cover may also be formed substantially of a polycarbonate copolymer resin.

(119) The parking meter **900** provides a low-profile parking meter assembly **901** that can be secured to a coin vault assembly, or similar structure. The parking meter assembly comprises an input area **906** comprising a plurality of input controls **902** arranged horizontally below a display **904**. It will be appreciated that other arrangements of input controls and/or displays are possible. Further, in order to reduce an overall height of the parking meter assembly **901**, the payment means are all located in the same general horizontal area located below the input area **906**. That is, the coin chute **908** is located horizontally adjacent to the card slot **910** for the card reader as well as a tap target, or antenna location, **912** for the contactless payment reader. As depicted in FIG. **9**, the payment means **908**, **910** and **912** are generally horizontally aligned with each other. This is in contrast to the coin chute **318** of parking meter **300**, which is vertically offset from, and so not generally horizontally

aligned with the other payment means **320, 322**. The parking meter **900** may further comprise an overhead light **914** projecting downwards from the front cover in order to illuminate at least a portion of the parking meter to facilitate user interactions in the evening or during dark periods. The parking meter may include a solar panel **916** located at a top section of the parking meter. As described further below, the parking meter assembly **901** may include an internal structure to which various parking meter components may be connected. A front housing cover **918** may be secured to the internal structure, and the internal structure and front housing **918** of the parking meter assembly **901** may slide up and down a rear assembly **920** that remains fixed to the coin vault.

(120) FIG. **10** depicts a rear isometric view of the single space parking meter of FIG. **9** with the parking meter assembly partially raised. As depicted, a rear assembly **920** can be secured to the coin vault **102** and saddle plate **922** that may be secured over the coin vault **102**. The rear assembly **920** may comprise a bottom plate **924** that can be secured to the saddle plate **922** and to the coin vault housing **102** using bolts or other connection means. A rear cover **989** may be secured to the bottom plate **924** using any number of fastening techniques such as welding, screws, bolts, etc. A locking assembly may comprise a locking component for preventing vertical movement of the front assembly **901** relative to the rear assembly **920** when engaged with the front assembly in a lowered position. When the vertical locking component is disengaged, the front assembly **901** may be slid up and/or down the rear assembly **920**. The front assembly **901** may be partially slid up the rear assembly **920** to a service position as depicted in FIG. **10** that allows for inspection and servicing of internal parking meter components. Although described as sliding vertically to a service position, it is possible for other movements to be provided. For example, it is possible for the front assembly **926** to slide vertically and then rotate or hinge so as to place the front assembly in an angled or at least partially horizontal position to facilitate servicing and inspections. The front assembly **901** may be completely removed from the rear assembly **920** for replacement or other purposes.

(121) As depicted in FIG. **10**, the front assembly **901** may comprise the front cover **918** which may be securely attached to an internal structure **928**. The internal structure **928** and/or the front cover **918** may provide mounting connections for securing various parking meter components to the front assembly **901**. The components may be secured using a quick-connection type of mechanism that allows individual components to be moved for inspection and/or removed for servicing or replacing without requiring any tools for such moving and/or removal. The components may include a solar panel assembly or module that includes a solar panel **916**, a radio board component or radio antenna compartment **930**, a battery compartment **932** (depicted with a battery), a magnetic stripe and/or contact based card reader **934**, a coin chute **936** and/or a contactless payment card reader and antenna **938**.

(122) FIG. **11** depicts an internal isometric view of a front cover of the single space parking meter of FIG. **9**. The front cover **918** may be formed at least in part from a polycarbonate copolymer resin to provide desirable protection against vandalism and the environment. Although formed at least in part from a polycarbonate copolymer resin, the front cover **918** may include metal reinforcing components in potentially weak areas. For example, a metal reinforcing plate may be secured to the polycarbonate copolymer resin front cover **918** in the area surrounding a display opening. Other metal reinforcing structures or components may be provided at various locations where the polycarbonate copolymer resin front cover **918** may require additional mechanical strength. As depicted, the front cover **918** may provide various mounting structures for securing various components, such as displays, main circuit boards, input components, the internal structure, etc. to the front cover **918**. The front cover **918** includes an upper portion **940** where the solar panel assembly may be received. The upper portion may include a downward protrusion **942** above a user interface panel to provide a housing for an LED to illuminate the user interface panel. The upper portion may also include a ridge **944** for securely capturing the solar panel assembly when slid into the upper portion **940**. The front cover **918** may include two side edge surfaces **946, 948** that provide a bearing surface that in conjunction with bearing surfaces of the internal structure **928** capture rails of the rear assembly **920** and allow the front assembly **901** to slide up and down the rear assembly **920**.

(123) FIG. **12** depicts a rear isometric view of an internal structure **928** of the single space parking meter of FIG. **9**. The internal structure may be formed at least in part from the same polycarbonate co-polymer resin as the front cover **918**. However, since the internal structure **928** will be protected by the front cover **918**, and rear cover **989**, it may not be necessary to form the internal structure from the same material. Rather, the internal structure may be formed from other plastics. The internal structure may include various mounting points for securing components to the parking meter. As described above, individual components may be connected using quick-connect connections such as thumbscrews or similar friction type connections to secure the components in place. The internal structure **928** comprises a number of square openings **950** that may receive capturing components that allow, for example, a thumbscrew to secure the component with a quarter turn. The internal structure **928** includes two side edge surfaces **952**, only one of which is labeled, that provide bearing surfaces that cooperate with the bearing surfaces **946, 948** of the front cover to capture rails of the rear assembly **920**. The internal structure **928** may also include an opening or section **954** for receiving a locking component used to secure the front assembly from vertical movement. A locking bar or rod **956** may be secured within the internal structure **928** and may be captured by a hook (not shown) when the lock is engaged.

(124) FIG. **13** depicts a side view of the internal structure of FIG. **12** attached to the front cover of FIG. **11**. As depicted, when the front cover **918** is secured to the internal structure **928**, the corresponding bearing surfaces **946, 952** of the front cover **918** and the internal structure **928**, respectively, are in close proximity to each other forming a channel **958**. The channel **958** captures a rail of the rear assembly **920** to secure the front assembly **901** to the rear assembly **920** while allowing the front assembly **901** to slide up and down the rear assembly **920**.

(125) FIG. **14** depicts a top perspective view of the front cover **918** and removable components of the parking meter of FIG. **9**. As described above, the internal structure **928** (not shown in FIG. **14**) and/or the front cover **918** may provide mounting features to allow the payment components, such as coin chute **936**, card reader **934** and contactless payment reader **938** to be

releasably secured within the parking meter. As an example, the payment components may be secured to the front cover **918** and internal structure **928** using thumbscrews **960**. The thumbscrews **960** allow the components to be easily disengaged from the internal structure and/or front cover for inspection, service, or removal from the internal structure and/or front cover if necessary. The thumbscrews **960** may be inserted through a capturing component on the internal structure and turned an amount, such as a quarter turn, in order to secure the connection. For example, during routine maintenance on the street, a coin chute **936** may be quickly replaced with another coin chute by unscrewing the thumbscrew or thumbscrews and removing the coin chute **936**. Additionally or alternatively, unscrewing the thumbscrew may allow the payment component to be moved to a partially attached position which allows the payment component to be easily inspected while not being fully separated from the parking meter. For example, the payment components may be pivotally connected to the internal structure and/or front cover. Unscrewing the thumbscrew may allow the payment component to become partially disengaged from the internal structure and/or front cover and swing outwards into a disengagement position. Such a disengagement position may provide improved visibility for inspection and improved access for service without actual removal of the payment component from the parking meter. Once inspection or service has been completed, the payment component may be swung or rotated back to a fully engaged position and the thumbscrew re-attached to secure the payment device in the fully engaged position. If upon inspection or servicing it is determined that the payment component should be replaced, it can be removed from the disengagement position and a replacement component secured to the parking meter.

(126) FIG. **15** depicts a detail perspective view of a removable contactless payment reader **938**. The removable contactless payment reader **938** as depicted may include electronic components, such as the reader circuitry and antenna secured to, or formed on, one or more boards **962** that can be secured to a frame **964**. The frame **964** includes a thumbscrew **960** that can be used to secure the removable contactless payment reader **938** in place. The frame **964** includes a lower portion that has a protruding cylindrical portion **966** that may act as a pivot point of the connection.

(127) FIG. **16** depicts details of the mounting of the removable contactless payment reader of FIG. **15** to the internal structure of FIG. **12**. The contactless payment reader **938** is secured within the parking meter by a thumbscrew **960** that secures the reader **938** to the internal structure **928**. Additionally the cylindrical protrusion **966** of the frame **964** provides a second connection point that can help provide a secure connection of the removable contactless payment reader to the parking meter. The cylindrical protrusion may be partially secured against a retaining bar **968** that may be secured to the parking meter, for example by screws or other attachment means. When the internal structure **928**, and front cover (not shown) are raised, the removable contactless payment reader **938** may be removed by disengaging the thumbscrew **960** and lowering the removable contactless payment reader **938** so that the cylindrical protrusion clears the retaining bar **968** and the reader **938** can be fully removed from the parking meter. The contactless payment reader **938** may be partially disengaged from the internal structure by disengaging the thumbscrew **960**, but maintaining contact between the cylindrical protrusion **966** and the retaining bar **968**. Such partial disengagement may allow the component to be rotated for quick inspection. The component may then be easily re-secured to the internal structure **918** by the thumbscrew **960**.

(128) FIG. **17** depicts a side view of the front cover, internal structure and solar panel assembly. The parking meter may include a solar panel that is located in a solar panel assembly **970** that can be inserted into the front assembly **901**, and further electrically engages with a battery of the parking meter when the solar panel assembly is fully engaged with the front assembly. This allows the solar panel assembly **970** to be easily replaceable and/or serviceable in the event of vandalism, failure, or other damage or malfunction. The solar panel assembly **970** may include the one or more lights **972** that can be exposed through a small opening or downward protrusion **942** in the front cover **918** for illuminating the user interface of the parking meter. The solar panel assembly **970** may further include status LEDs or holes **974** for permitting the visibility of such status LEDs. As depicted, the solar panel assembly **970** may include a channel **976** that matches with the ridge **944** to secure the solar panel assembly **970** within the front assembly **901**. FIG. **17** also depicts a rotatable battery compartment **932** as well as the card payment reader **934**.

(129) FIG. **18** depicts a side view of the front assembly and radio assembly (or radio communication component). In addition to the slidable solar panel assembly **970** described above, the radio antenna of the parking meter may be provided in a radio antenna compartment or removable radio module **930** that slidably engages with or is removably coupled or secured to the front assembly **901** and further electrically engages with the main electronics board when it is fully engaged onto the front assembly. The radio antenna compartment **930** may be secured in position by, for example, thumbscrews **960**. The radio antenna compartment **930** comprises a housing **978** that can enclose radio electronics such as radio transceivers. An electrical connector **984** may be secured to the housing **978** so that when the radio antenna compartment is slid into the front assembly, the electrical connector mates with a corresponding connection to establish power and data connections. This allows the radio and antenna combination to be quickly and easily swapped out in the field for maintenance/service, to replace/inspect a SIM card, or to switch the meter from one cellular network to another alternative network that may have better coverage or signal strength than the original network. The radio antenna compartment **930** may include a metal ground plane **980** above which an antenna board **982** may be located. As will be appreciated, the metal ground plane may be used to provide desired radio frequency (RF) transmission performance from the antenna of the antenna board **982**. The radio antenna compartment **930** may also include conductive sprays applied onto inside surfaces of the compartment for the purpose of shielding the radio and reducing spurious emissions. As depicted in FIG. **19**, the radio antenna board **982** (as seen from above) may be contoured in order to follow the shape of the front cover **918** in order to locate the antenna in a favorable location adjacent the front cover.

(130) FIG. **20** depicts a rear perspective view of the front cover, internal structure and locking assembly with the rear cover removed for visibility. The locking assembly may comprise a cylindrical locking component **984** that is secured to the bottom plate **924**. Upon rotation of the cylindrical lock, a hook bar **986** rotates so that a hook portion captures a locking rod **956** fixed

within the front assembly **901**. By rotating the cylindrical lock **984** to a locked position with the hook bar engaged with the locking rod **956**, the front assembly cannot be raised from the bottom plate, which in turn is secured to the coin vault assembly.

(131) FIG. **21** depicts a perspective view of the rear assembly. The rear assembly **920** includes a rear cover **989** that provides the rear portion of the protective housing. The rear cover **989** may be formed at least partially from a polycarbonate copolymer resin. Preferably, the rear cover is substantially formed from a polycarbonate copolymer resin. Alternatively, the rear cover **989** may be formed from metal, such as stainless steel, or other rugged materials to provide the desired protection against vandalism and the environment. The rear cover **989** includes two rails **990**, **992** that can be received within the channel **958** between the bearing surfaces of the front cover **918** and the internal structure **928**. The front assembly can be slid up and down the rails **990**, **992**. The rails may include an indent **994** that can engage a releasable locking pin to maintain the front assembly in a partially raised position, such as that of FIG. **10**, to facilitate servicing and inspections. The rear assembly **920** further includes the cylindrical lock **984** and attached hook bar **986** for securing the front assembly to the rear assembly and preventing vertical movement. The rear assembly may further include a second locking component **995** that prevents rotation of the rear assembly **920**. The rear assembly may be secured to the coin vault housing by twisting the rear cover, and attached bottom plate, so that expanded bolt heads secured to the coin vault housing are positioned above a reduced-sized opening. The bolt heads in this position, in combination with the reduced-sized openings of the bottom plate, prevent the rear assembly from being lifted away from the coin vault assembly. As a result, the combination of the second locking component **995** and bolt heads positioned above reduced-sized openings prevents withdrawal or removal of the rear assembly from the coin vault assembly and ensures the two assemblies are secured together. The second locking component or rotational lock **995** comprises a locking arm **996** pivotally connected to the bottom plate **924**. The locking arm **996** includes two protrusions **998** (only one is labeled), although other numbers of protrusions are possible, that when the locking arm is in a lowered locked position, protrude through openings in the bottom plate and are received in corresponding apertures in the coin vault assembly to prevent rotation. When in an unlocked position, the protrusions **998** do not extend into the apertures of the coin vault assembly, and the rear assembly can be rotated such that the securing bolt heads of the coin vault assembly may be positioned within the expanded portions of the bottom plate openings. As such, the securing bolt heads can be lifted or withdrawn through the openings. Alternatively, the rear assembly may be secured to the coin vault housing using other fastening techniques such as screws or bolts, wherein the bottom plate is not rotatable in the manner as described above.

(132) FIG. **22** depicts details of service locks. As described above with reference to FIG. **21**, the rails **990**, **992** upon which the front assembly slides up and down may include indents or notches **994**. Service locking bars **1008** may be pivotally connected to the internal structure **928**. The locking bars **1008** may include a biasing spring **2010** at one end of the locking bars **1008** to bias the locking bars **1008** into a locked position. The locked position forces the lower end of the locking bars which includes a foot protrusion **1012** into the rail's notch or indent as it slides past. The notch **994** and foot protrusions **1012** may have an angled surface to allow the foot protrusion to be forced out of the notch **994** when sliding the front assembly upwards. However, a bottom of the foot protrusions **1012** and notches may be relatively flat so that upon sliding the front assembly down the rails, the foot protrusions may engage with the notches **994** to prevent further sliding down of the front assembly. The locking bars **1008** may be manually disengaged from the notches **994** by pressing the spring biased portion inwards to release the foot protrusions **1012** from the notches. In addition, and as discussed in greater detail above, the front assembly may be raised to a first position such that rail guides at the top of the front assembly may be released from the rails of the rear assembly at this first position, while the bottom of the front assembly remains engaged with the rear assembly. Under this configuration, there would also be a hinge, such as a telescoping or folded metal rod, located at a release point, thus allowing movement or rotation of the front assembly so that even more convenient access to the internal components for servicing or removal is made possible.

(133) The above has described the use of polycarbonate copolymer resins in forming lightweight and vandal resistant parking meter housings. Although various polycarbonate copolymer resins may be used, one commercially available example is the Lexan EXL® resin series made by SABIC (Saudi Arabia Basic Industries Corporation). Lexan EXL® resin is a copolymer formulation of polycarbonate and polysiloxane providing a material having high impact resistance, low temperature ductility and favourable humidity characteristics. Polycarbonate copolymer resins such as the Lexan EXL® resin series may also include electrical and flame retardant properties, UV-stabilizers, improved aesthetics such as high surface finish, and a broad spectrum of both opaque and transparent colours. In addition, as a favourable plastic for addressing eco-footprint concerns and demands, the Lexan EXL® resin series is based on non-chlorine and non-bromine flame retardant systems, and is free from heavy-metal pigments. In addition to the Lexan EXL® resin series, and as discussed further below, an alternative polycarbonate copolymer resin, or polycarbonate blended resin that may be used is the Makroblend UT1018 resin made by Covestro LLC.

(134) In contrast to polycarbonate copolymer resins such as the Lexan EXL® resin series, non-copolymer or traditional polycarbonate material is less flexible, more susceptible to stress cracking or fracturing (particularly in cold conditions), more easily marked, has lower abrasion resistance, and has less favorable ductility and flexibility properties over a wide temperature range. In addition, non-copolymer or traditional polycarbonate is less favorable than nylon with respect to chemical resistance.

(135) In the alternative to a polycarbonate copolymer, or polycarbonate blended resin, such as the Lexan EXL® resin series or the Makroblend UT1018 resin, other commercially available example materials that may be used for the housing exist. For example, the OnForce™ M LFT NN-60LGF/000 HS Natural product of PolyOne Corporation, and the RTP 299 A×140812 A product of RTP Company, each when combined or included with long glass fiber (LGF) or Kevlar fiber, may be suitable materials for the parking meter housing.

(136) Both of the above-noted alternative products (of PolyOne Corporation and RTP Company) are nylon-based, and are technically known as aliphatic or semi-aromatic polyamides (PA). (The prefix “PA” and “nylon” are equivalent in meaning.) In similar impact tests, the Lexan EXL® was found to be preferable to both of the above-noted materials even when those materials were combined with LGF. The OnForce (PolyOne Corporation) product noted is PA 66 made from hexamethylenediamine and adipic acid. The RTP product is PA 6 made from ε-Caprolactam. Both products are considered homopolymers. Copolymers, on the other hand, comprise a plurality of monomers, wherein one monomer of a copolymer is known as a comonomer. Both the OnForce and RTP products may preferably be combined with or include long glass fiber (LGF) material to increase their strength. Including LGF, or alternatively, Kevlar fiber, in this regard, results in a preferable material with respect to strength and other desirable characteristics as compared to standard nylon, such as standard PA 66 or standard PA 6.

(137) PA 6 and PA 66 nylon materials can alternatively be combined to form a copolymer, i.e., PA 6/66 made from caprolactam, hexamethylenediamine and adipic acid. As they stand on their own, they have not been mixed or combined, and as such their crystallinity (which contributes strength and rigidity) and melting point are typically greater than the crystallinity and melting point of a copolymer that has resulted from a mixture of comonomers or pairs of comonomers. However, a mixture that forms a copolymer may still possess high crystallinity while increasing the amorphous regions in a material which contributes to its flexibility or elasticity. An example of a PA 6/66 copolymer that is nylon-based and that may be used to substantially or significantly form a parking meter housing is as DuPont's ZYTEL® 74G33J.

(138) Preferably, a plastic copolymer or blended resin material that strikes a balance between crystallinity and elasticity (i.e., stiffness and toughness) is used. Standard or conventional nylon and polycarbonate materials do not offer the desired balance that achieves the materials characteristics found in a copolymer or a blended resin, such as those described herein. Moreover, standard or conventional nylon and polycarbonate materials may not provide material characteristics or performance that would make them preferable or acceptable for use or deployment in applications previously addressed by metal, such as the external housing of a parking meter, when compared with the material characteristics and performance of a copolymer or blended resin. In the alternative to a copolymer or blended resin, a nylon based polymer that includes long glass fiber or Kevlar fiber may be preferable over standard or conventional nylon and polycarbonate materials.

(139) As noted above, Lexan EXL® is a copolymer made of polycarbonate and siloxane (silicone rubber). Nylon-based materials tend to absorb moisture at a rate greater than other plastics such as polycarbonate. Polycarbonate materials have a lower RF absorption coefficient (wherein a lower coefficient makes for better RF performance) than nylon-based materials. Unlike most thermoplastics, polycarbonate can undergo significant plastic deformations without cracking or breaking. A polycarbonate blend that may be an alternative to the Lexan EXL® polycarbonate copolymer and that may be used to substantially or significantly form a parking meter housing is the Makroblend UT1018 resin made by Covestro LLC, which is an impact-modified blend of Makrolon® polycarbonate and polyethylene terephthalate (PET) polyester. This plastic blend combines the physical toughness of polycarbonate with the favorable chemical resistance of PET, and provides favorable chemical resistance, favorable low-temperature impact strength, and favorable dimensional stability, even in high-moisture environments. Its fatigue endurance and abrasion resistance are preferable to that of standard polycarbonate. The naturally glossy finish of Makroblend UT1018 resin provides an attractive surface that does not require painting. The resin is opaque and is supplied in pellet form for injection molding or extrusion.

(140) The above-described favorable characteristics of polycarbonate copolymer and polycarbonate blended resins, nylon-based copolymers, and nylon-based polymers that include long glass fiber or Kevlar fiber are not commonly or generally known to those skilled in the art of parking meters. In this regard, the common general knowledge of those skilled in the art of parking meters is to prefer metallic parking meter housings given such housings are widely accepted and in use based on a strong track record for security and vandal-resistance, given use of plastics in parking meters has been overwhelmingly limited to internal molded plastic components and limited external components such as clear plastics for windows that cover a user display, and/or given the disadvantages of traditional polycarbonate (i.e., non-copolymer) or other traditional plastic material that would or may be felt if such material were used in or for a parking meter housing.

(141) Although various individual features and/or functionality may have been described with reference to a specific embodiment, such features and/or functionality may be incorporated into other embodiments.

(142) Although specific embodiments of a retrofit parking meter are described, it will be appreciated that other parking meter designs, including other parking meter components or configurations of parking meter components, may be implemented. Variations and modifications not described herein will be apparent to one of ordinary skill in the art having regard to the description herein.

Claims

1. A meter comprising: a first portion of an external housing having a bottom plate secured at a location to a top of an existing structure, a cross-section of the top of the existing structure having a first profile shape, the first portion of the external housing defining at least a pair of engageable rails above the bottom plate; and a removable assembly comprising: one or more meter payment components comprising one or more of a card payment reader, contactless payment reader and a coin payment reader; a radio frequency (RF) radio for wireless communication; and a second portion of an external housing comprising a pair of co-operating rails; wherein the pair of co-operating rails of the second portion of the external housing are engageable with the pair of engageable rails of the first portion of the external housing to allow relative movement of the first portion of the external housing and the second portion of the external housing between a first orientation of the removable assembly in which the removable assembly is detached from the first portion of the external housing and a second orientation

of the removable assembly in which at least the first portion of the external housing and the second portion of the external housing provide a unitary housing to provide protection for the RF radio and the one or more meter payment components against an external environment and vandalism attack, a cross-section of the unitary housing at the bottom plate having a second profile shape, the unitary housing providing a transition portion extending from the bottom plate to provide a transition between the first profile shape of the top of the existing structure and the second profile shape of the bottom of the unitary housing.

2. The meter of claim 1, wherein at least one of the first portion of the external housing and the second portion of the external housing are formed at least partially from a material having a low attenuation of RF signals.
3. The meter of claim 2, wherein the RF radio for wireless communication comprises one or more of: a near-field communication (NFC) reader; a Bluetooth radio; a ZigBee radio; a WiFi radio; and a cellular radio.
4. The meter of claim 2, wherein the material having a low attenuation of RF signals comprises an impact resistant plastic.
5. The meter of claim 4, wherein the impact resistant plastic comprises at least one of a polycarbonate copolymer, a polycarbonate blended resin, and a nylon-based copolymer.
6. The meter of claim 4, wherein the second portion of the external housing is formed substantially of the impact resistant plastic.
7. The meter of claim 6, wherein the first portion of the external housing is formed from metal.
8. The meter of claim 1, wherein the bottom plate is formed from metal.
9. The meter of claim 1, wherein the transition portion of the unitary housing is formed from an impact resistant plastic.
10. The meter of claim 1, wherein the existing structure comprises a coin vault.
11. The meter of claim 10, wherein the first profile shape is rectangular.
12. The meter of claim 11, wherein the second profile shape is circular.
13. The meter of claim 1, wherein the transition portion of the unitary housing extends down below the bottom plate.
14. The meter of claim 1, wherein the second portion of the external housing comprises a flat input area comprising at least one input control.
15. The meter of claim 14, wherein the payment components are arranged generally horizontally on the flat input area.
16. The meter of claim 1, wherein the first portion of the external housing is lockably securable to the second portion of the external housing, and at least some meter components are secured to the second portion of the external housing.
17. The meter of claim 1, wherein the unitary housing is generally cylindrical in shape, and wherein the second portion of the external housing slides vertically relative to the first portion of the external housing.
18. The meter of claim 1, wherein the removable assembly is moveable to a third orientation in which the removable assembly is at least partially supported by the rails of the first portion of the external housing to allow access to one or more meter components.
19. The meter of claim 18, wherein at least one of the pair of rails and pair of co-operating rails comprises a spring-catch assembly to releasably support the second portion of the external housing on the first portion of the external housing with the removable assembly in the third orientation.
20. The meter of claim 1, wherein the meter is a single space parking meter.
